

1. Administrative & Study Identification

Full Study Title: A Phase III, Randomized, Double-blind, Placebo-controlled Study Evaluating Atezolizumab and Bevacizumab, With or Without Tiragolumab, in Patients With Untreated Locally Advanced or Metastatic Hepatocellular Carcinoma **Short Title/Acronym:** IMbrave152 / SKYSCRAPER-14 **Protocol Number:** C044668 **NCT Number:** NCT05904886 **Sponsor Name:** Hoffmann-La Roche (Roche)
Investigational Product: Tiragolumab, Atezolizumab, Bevacizumab (Trade names include: Avastin, Vegzelma, Alymsys, Zirabev, Mvasi)
Phase of Development: Phase III **Trial Status:** Active, not recruiting **Medical Monitor:** Hoffmann-La Roche Medical Monitor

2. Study Synopsis & Rationale

2.1 Study Objective & Summary Primary Objective: To assess the efficacy and safety of tiragolumab, an anti-TIGIT monoclonal antibody, when administered in combination with atezolizumab and bevacizumab as first-line treatment in participants with unresectable, locally advanced or metastatic hepatocellular carcinoma (HCC). **Recent Amendment Update (Version 5):** Following a memo issued by the Sponsor, participants receiving treatment in the atezolizumab plus bevacizumab plus tiragolumab arm are recommended to discontinue tiragolumab treatment unless the investigator decides the benefit outweighs the risk. Participants receiving treatment in the atezolizumab plus bevacizumab plus placebo arm must discontinue placebo treatment. Participants may continue receiving active treatment(s) per protocol until loss of clinical benefit or unacceptable toxicity, whichever occurs first.

2.2 Background & Rationale Atezolizumab plus bevacizumab is a standard of care, first-line therapy for advanced HCC. Myeloid and T regulatory cells are key immunosuppressive cell types within the hepatic tumor microenvironment associated with clinical resistance to atezolizumab and bevacizumab therapy and overall poor prognosis. Therapeutic targeting of TIGIT, which is highly expressed in these cells, with tiragolumab may overcome the immunosuppressive environment and improve clinical benefit.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind **Randomization:** Randomized

3.2 Planned Enrollment & Duration Target **\$N\$:** 687 participants
Number of Sites: Global recruitment (including Asia, Africa, USA, etc.) **Estimated Study Start:** 14-Sep-2023 **Estimated Primary Completion:** TBD

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years. 2. Locally advanced or metastatic and/or unresectable HCC with diagnosis confirmed by histology/cytology or clinically by American Association for the Study of Liver Diseases (AASLD) criteria in cirrhotic participants. 3. Disease that is not amenable to curative surgical and/or locoregional therapies. 4. No prior systemic treatment for locally advanced or metastatic and/or unresectable HCC-measurable disease according to RECIST v1.1. 5. Eastern Cooperative Oncology Group (ECOG) performance status of 0 or 1 within 7 days prior to randomization. 6. Child-Pugh Class A within 7 days prior to randomization. 7. Adequate hematologic and end-organ function. 8. Female participants of childbearing potential must be willing to avoid pregnancy within 5 months after the final dose of atezolizumab, within 6 months after the final dose of bevacizumab, and within 90 days after the final dose of tiragolumab/placebo. 9. Male participants with a female partner of childbearing potential or pregnant female partner must remain abstinent or use a condom during the treatment period and for 6 months after the final dose of bevacizumab and for 90 days after the final dose of tiragolumab/placebo to avoid exposing the embryo.

4.2 Exclusion Criteria 1. Pregnancy or breastfeeding within 5 months after the final dose of atezolizumab, within 6 months after the final dose of bevacizumab, and within 90 days after the final dose of tiragolumab/placebo. 2. Prior treatment with cluster of differentiation 137 (CD137) agonists or immune checkpoint blockade therapies. 3. Treatment with investigational therapy within 28 days prior to initiation of study treatment. 4. Treatment with locoregional therapy to liver within 28 days prior to initiation of study treatment, or non-recovery from side effects of any such procedure. 5. Treatment with systemic immunostimulatory agents or systemic immunosuppressive medication. 6. Untreated or incompletely treated esophageal and/or gastric varices with bleeding or that are at high risk for bleeding. 7. A prior bleeding event due to esophageal and/or gastric varices within 6 months prior to initiation of study treatment. 8. Active or history of autoimmune disease or immune deficiency. 9. History of idiopathic pulmonary fibrosis, organizing pneumonia, drug-induced pneumonitis, or idiopathic pneumonitis, or evidence of active pneumonitis on screening chest computed tomography (CT) scan. 10. History of malignancy other than HCC within 5 years prior to screening, with the exception of malignancies with a negligible risk of metastasis or death. 11. Mixed histology or other subtypes/variants of HCC, including, but

not limited to, known liver adenocarcinoma, fibrolamellar HCC, sarcomatoid HCC, other rare HCC variant, or mixed cholangiocarcinoma and HCC. 12. Co-infection with hepatitis B virus (HBV) and hepatitis C virus (HCV). 13. Acute Epstein-Barr virus (EBV) infection or known or suspected chronic active EBV infection. 14. Symptomatic, untreated, or actively progressing central nervous system (CNS) metastases.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Tiragolumab (or placebo) plus atezolizumab plus bevacizumab administered approximately every 3 weeks. (*Note: Per amendment version 5, tiragolumab is recommended to be discontinued unless investigator decides benefit outweighs risk, and placebo must be discontinued*). **Route of Administration:** Intravenous (IV). **Duration of Treatment:** Continued until loss of clinical benefit or unacceptable toxicity, whichever occurs first.

5.2 Concomitant & Prohibited Therapy Permitted: Standard of care therapies for co-morbidities. **Prohibited:** Prior systemic treatment for locally advanced or metastatic HCC, CD137 agonists, systemic immunostimulatory agents, or systemic immunosuppressive medication.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
1	-28 to 0	Informed Consent, Medical History, Tumor Imaging, Eligibility Checks
2	0	Randomization, First Dose
3	Every 3 Weeks	IV Infusions, Safety Assessments, Efficacy Assessment (Tumor Scans)
4	Upon progression/toxicity	End of Treatment Visit, Follow-up for Survival Data

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints 1. **Investigator-assessed Progression-Free Survival (PFS):** According to Response Evaluation Criteria in Solid Tumors Version 1.1 (RECIST v1.1). 2. **Overall Survival (OS):** The length of time participants live.

7.2 Secondary & Exploratory Endpoints 1. **Investigator-assessed Confirmed Objective Response Rate (ORR):** According to RECIST v1.1. 2. **Investigator-assessed Duration of Response (DOR):** According to RECIST v1.1. 3. **Investigator-assessed PFS Rate:** According to RECIST v1.1 at 6 and 12 Months.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Routine safety assessments performed regularly during hospital visits (about every 3 weeks). **Serious Adverse Events (SAEs):** Reported within standard Phase 3 global oncology trial timelines (typically 24 hours of site awareness). **Data Monitoring Committee (DMC):** Standard internal/external data monitoring committees implemented.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for primary efficacy outcomes. **Sample Size Justification:** Target enrollment of 687 patients to ensure adequate statistical power for evaluating differences in PFS and OS. **Handling of Missing Data:** Standard right-censoring rules utilized for time-to-event endpoints (PFS and OS).

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Regular onsite and remote monitoring of trial sites. **Audit Trail:** Data tracking, eCRF locking, and reporting maintained securely per sponsor protocols.

11. Study Identifiers & Governance

Primary ID: C044668 **Registry Number:** NCT05904886 **Sponsor/Lead Organization:** Hoffmann-La Roche (Roche) **Study Title:** IMbrave152 / SKYSCRAPER-14 **Official Regulatory Title:** A Phase III, Randomized, Double-blind, Placebo-controlled Study Evaluating Atezolizumab and Bevacizumab, With or Without Tiragolumab, in Patients With Untreated Locally Advanced or Metastatic Hepatocellular Carcinoma

12. Clinical Framework (Hepatocellular Carcinoma Specific)

Primary Condition: Hepatocellular Carcinoma (HCC) **Diagnosis Threshold:** Untreated locally advanced or metastatic HCC **Medical Methodology:** Anti-PD-L1, Anti-VEGF, and Anti-TIGIT Monoclonal Antibody combination therapy **Optimization Target:** Progression-Free Survival and Overall Survival

13. Study Procedures & Methodology

Management Pathway: Standard first-line systemic therapy clinical pathway for advanced HCC **Education Protocol (HCP):** Protocol administration training and AE management guidelines **Education Protocol (Patient):** Informed consent, potential side effects counseling, and hospital visit requirements **Quality Audit Mechanism:** Efficacy and radiologic scan reviews, safety profile monitoring

14. Observation & Follow-Up Schedule

Total Duration: Follow-up until loss of clinical benefit, death, or study termination **Onsite Visit Cadence:** Approximately every 3 weeks **Remote Monitoring Frequency:** As required between dosing cycles **Checklist Review Interval:** Every 3 weeks prior to drug administration

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---	
Principal Investigator	[Name]	_____	07-Apr-2026					
Clinical Operations Lead	[Name]	_____	07-Apr-2026		Lead			
Statistician	[Name]	_____	07-Apr-2026					